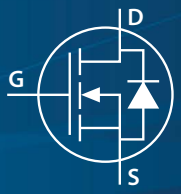
EPC2308 – Enhancement Mode Power Transistor

V_{DS} , 150 V
$R_{DS(on)}$, 6 mΩ max

PRELIMINARY



EPC EFFICIENT POWER CONVERSION

RoHS Halogen-Free

Gallium Nitride's exceptionally high electron mobility and low temperature coefficient allows very low $R_{DS(on)}$, while its lateral device structure and majority carrier diode provide exceptionally low Q_G and zero Q_{RR}. The end result is a device that can handle tasks where very high switching frequency, and low on-time are beneficial as well as those where on-state losses dominate.

Application Notes:

- Easy-to-use and reliable gate, Gate Drive ON = 5 V typical, OFF = 0 V (negative voltage not needed)
- Top of FET is electrically connected to source

Questions:

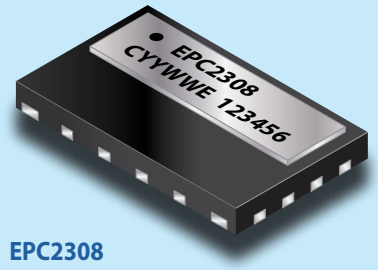
ASK AN EXPERT GaN

Maximum Ratings			
PARAMETER		**VALUE**	**UNIT**
V_{DS}	Drain-to-Source Voltage (Continuous)	150	V
	Drain-to-Source Voltage (up to 10,000 5 ms pulses at 150°C)	180	
I_D	Continuous (T_A = 25°C)	48	A
	Pulsed (25°C, T_{PULSE} = 300 µs)	157	
V_{GS}	Gate-to-Source Voltage	6	V
	Gate-to-Source Voltage	-4	
T_J	Operating Temperature	-40 to 150	°C
T_{STG}	Storage Temperature	-40 to 150	

Thermal Characteristics			
PARAMETER		**TYP**	**UNIT**
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case (Case TOP)	0.5	°C/W
$R_{\theta JB}$	Thermal Resistance, Junction-to-Board (Case BOTTOM)	2.8	
$R_{\theta JA_JEDEC}$	Thermal Resistance, Junction-to-Ambient (using JEDEC 51-2 PCB)	54	
$R_{\theta JA_EVB}$	Thermal Resistance, Junction-to-Ambient (EPC90143 EVB)	23	

Static Characteristics (T_J = 25°C unless otherwise stated)						
PARAMETER		**TEST CONDITIONS**	**MIN**	**TYP**	**MAX**	**UNIT**
BV_{DSS}	Drain-to-Source Voltage	V_{GS} = 0 V, I_D = [TBD]	150			V
I_{DSS}	Drain-Source Leakage	V_{GS} = 0 V, V_{DS} = 120 V		0.003		mA
I_{GSS}	Gate-to-Source Forward Leakage	V_{GS} = 5 V		0.015		
	Gate-to-Source Forward Leakage#	V_{GS} = 5 V, T_J = 125°C		0.2		
	Gate-to-Source Reverse Leakage	V_{GS} = -4 V		0.015		
$V_{GS(TH)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 5 mA	0.7	1.2	2.5	V
$R_{DS(on)}$	Drain-Source On Resistance	V_{GS} = 5 V, I_D = 15 A		4.6	6	mΩ
V_{SD}	Source-Drain Forward Voltage#	I_S = 0.5 A, V_{GS} = 0 V		1.5		V

Defined by design. Not subject to production test.



EPC2308
Package size: 3 x 5 mm

Applications

- High density DC-DC from 80–100 V
- AC/DC
- Synchronous rectification from 28–54 V for chargers, adaptors, and power supplies
- Solar optimizers and microinverters
- Motor drive and DC-DC for battery-operated power tools and robots
- USB fast chargers

Benefits

- Higher Efficiency – Lower conduction and switching losses, zero reverse recovery losses
- Ultra Small Footprint – Higher power density
- Thermally enhanced QFN package with exposed top and ultra-low thermal resistances for cooler operations
- Wettable flanks and 0.6 mm between high voltage and low voltage pads to simplify assembly and inspection

Scan QR code or click link below for more information including reliability reports, device models, demo boards!

https://bit.ly/EPC2308

Dynamic Characteristics# (T_J = 25°C unless otherwise stated)						
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
C_{ISS}	Input Capacitance	V_{DS} = 75 V, V_{GS} = 0 V		1454	2103	pF
C_{RSS}	Reverse Transfer Capacitance			2.6		
C_{OSS}	Output Capacitance			405	592	
$C_{OSS(ER)}$	Effective Output Capacitance, Energy Related (Note 1)	V_{DS} = 0 to 75 V, V_{GS} = 0 V		498		
$C_{OSS(TR)}$	Effective Output Capacitance, Time Related (Note 2)			664		
R_G	Gate Resistance			0.4		Ω
Q_G	Total Gate Charge	V_{DS} = 75 V, V_{GS} = 5 V, I_D = 15 A		10.6	13.8	nC
Q_{GS}	Gate-to-Source Charge	V_{DS} = 75 V, I_D = 15 A		3.8		
Q_{GD}	Gate-to-Drain Charge			1.3		
$Q_{G(TH)}$	Gate Charge at Threshold			2.4		
Q_{OSS}	Output Charge	V_{DS} = 75 V, V_{GS} = 0 V		50	61	
Q_{RR}	Source-Drain Recovery Charge			0		

All measurements were done with substrate connected to source.
Defined by design. Not subject to production test.
Note 1: $C_{OSS(ER)}$ is a fixed capacitance that gives the same stored energy as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.
Note 2: $C_{OSS(TR)}$ is a fixed capacitance that gives the same charging time as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.

Figure 1: Typical Output Characteristics at 25°C

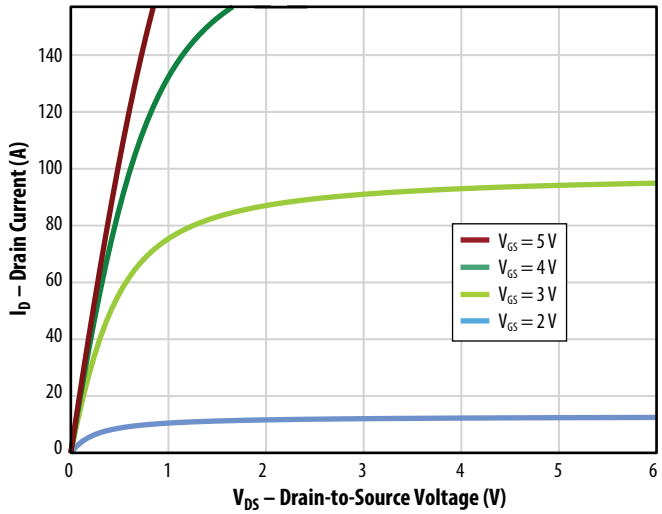


Figure 2: Typical Transfer Characteristics

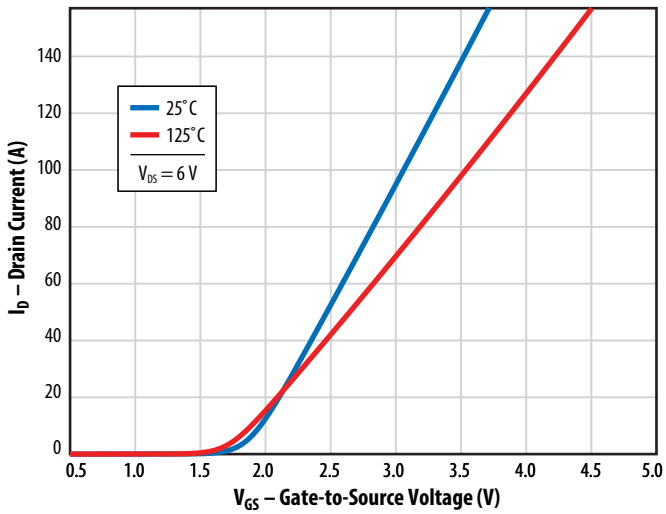


Figure 3: Typical $R_{DS(on)}$ vs. V_{GS} for Various Drain Currents

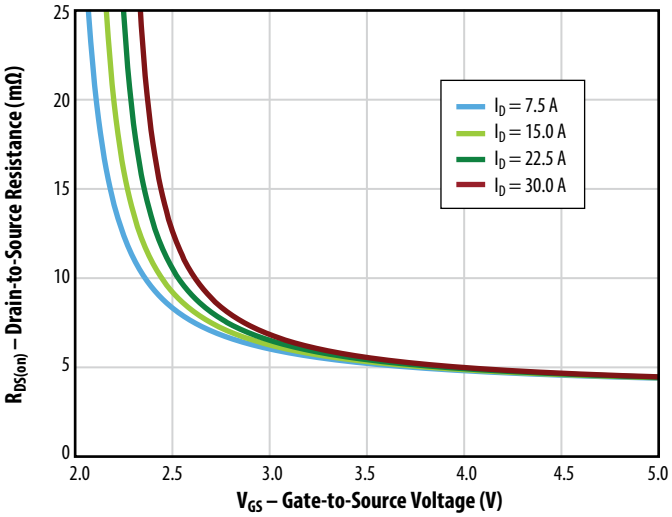


Figure 4: Typical $R_{DS(on)}$ vs. V_{GS} for Various Temperatures

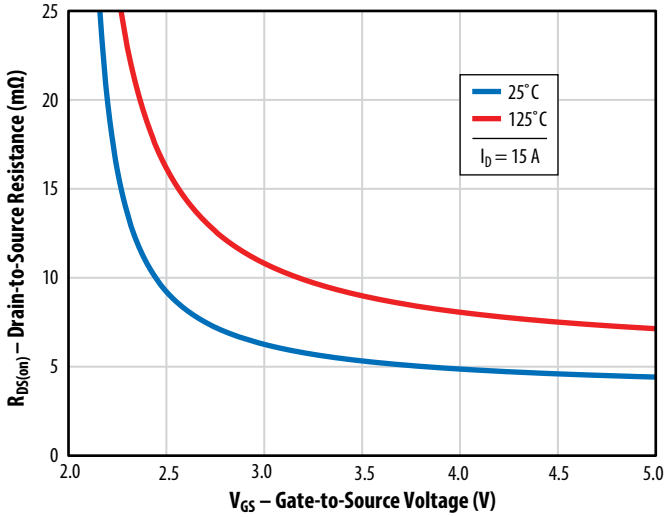


Figure 5a: Typical Capacitance (Linear Scale)

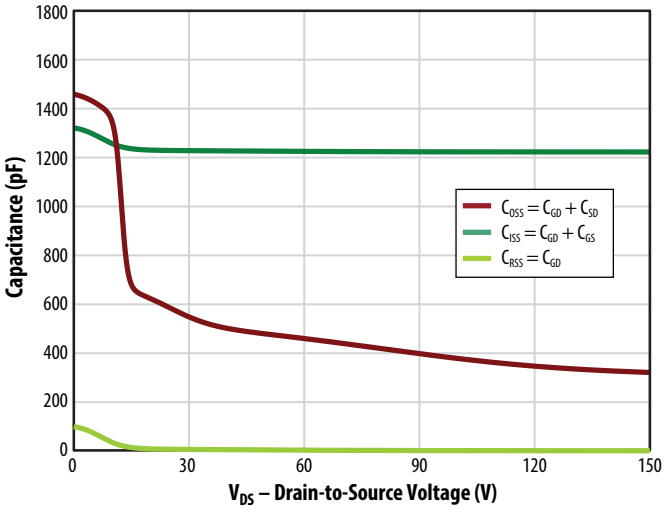


Figure 5b: Typical Capacitance (Log Scale)

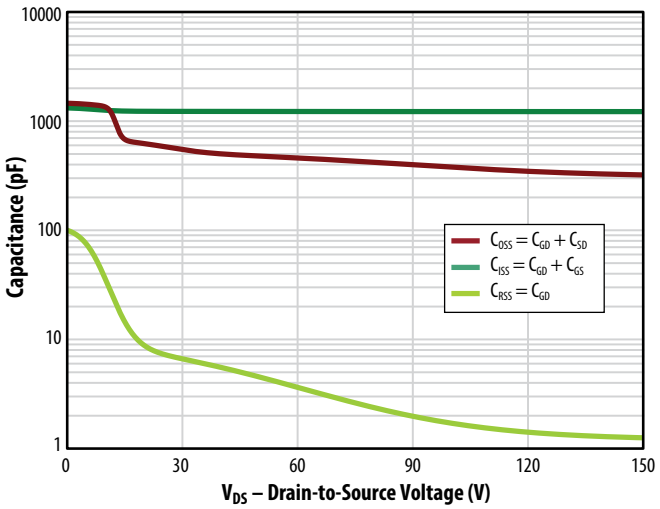


Figure 6: Typical Output Charge and C_{OSS} Stored Energy

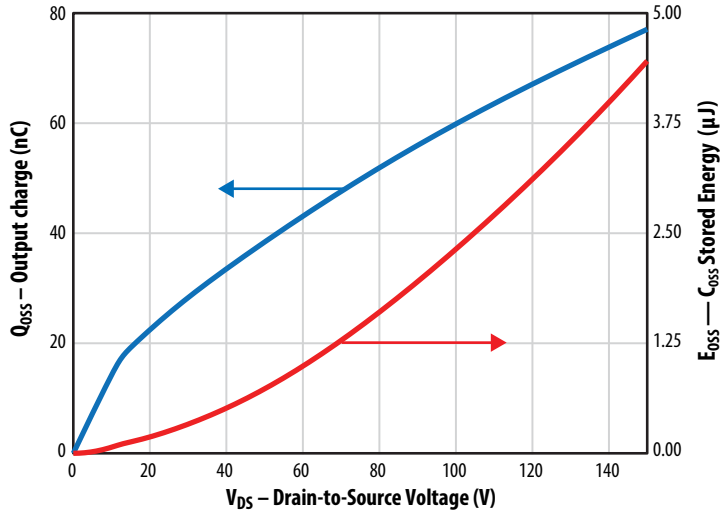


Figure 7: Typical Gate Charge

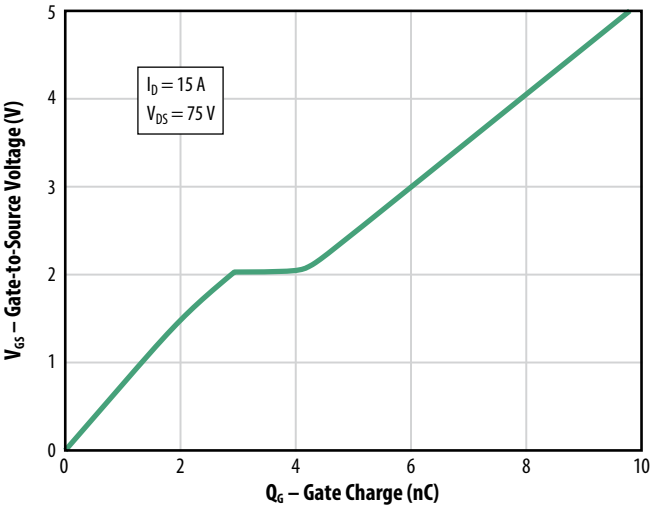
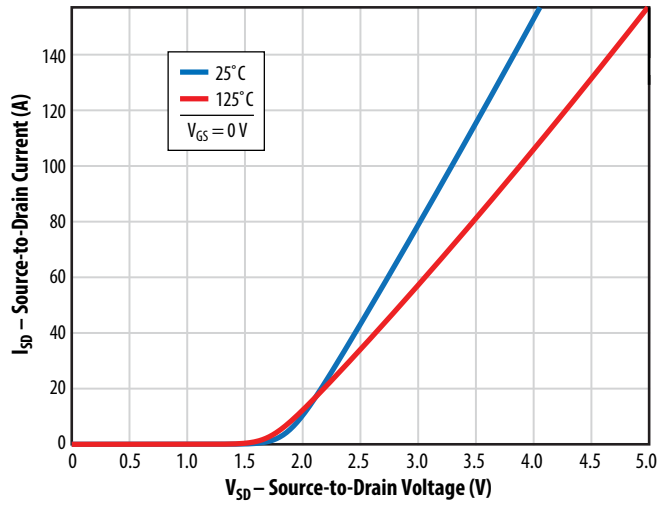


Figure 8: Typical Reverse Drain-Source Characteristics



Note: Negative gate drive voltage increases the reverse drain-source voltage.
EPC recommends 0 V for OFF.

Figure 9: Typical Normalized On-State Resistance vs. Temp.

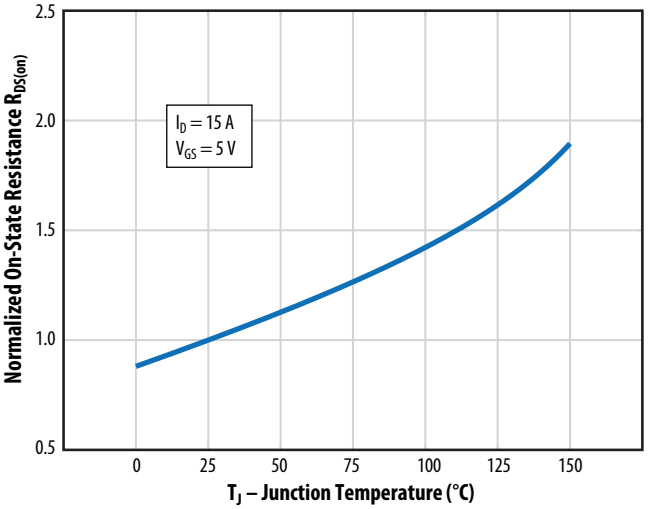


Figure 10: Typical Normalized Threshold Voltage vs. Temp.

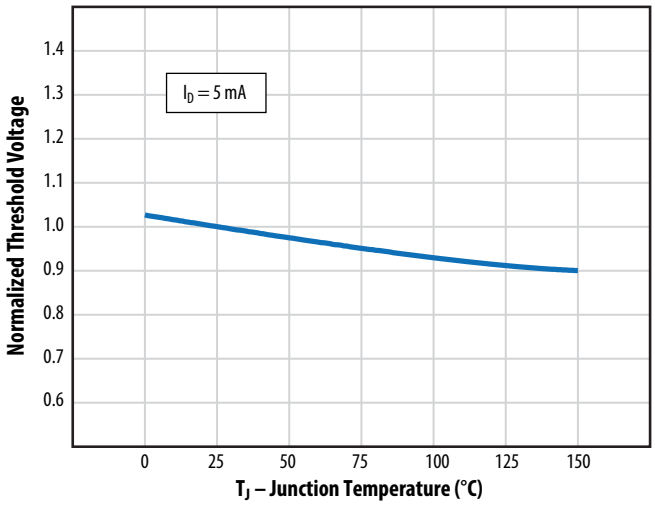


Figure 11: Safe Operating Area

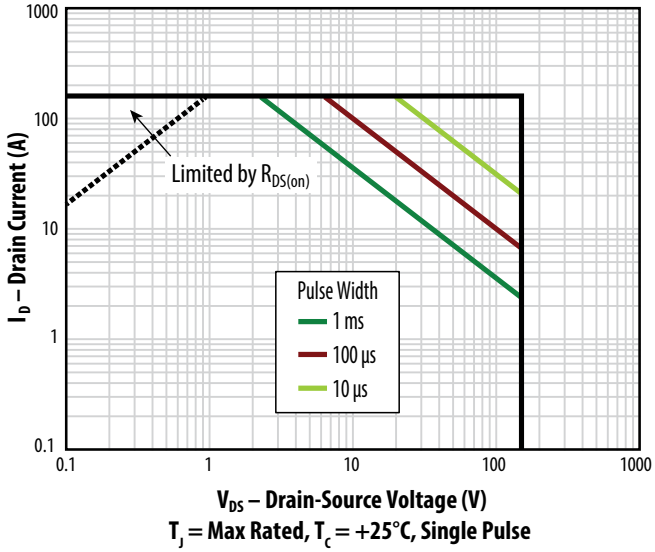
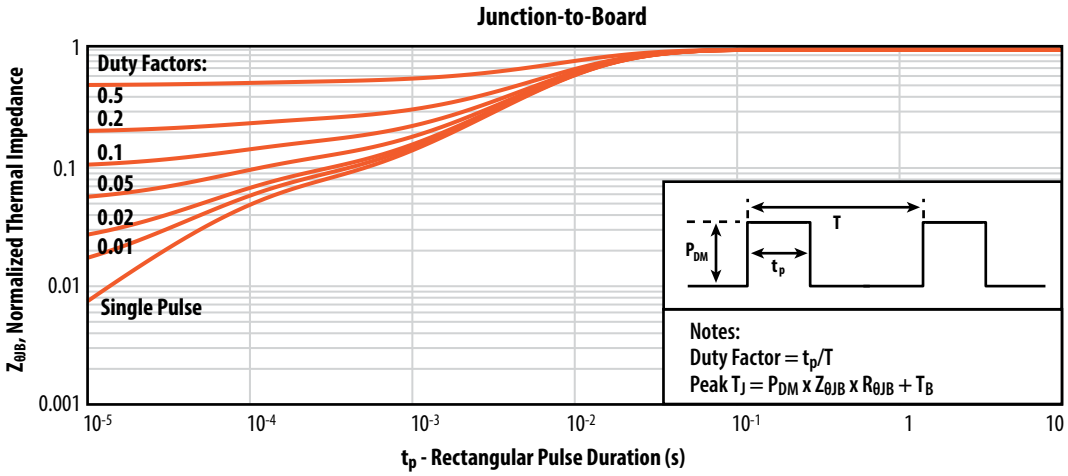
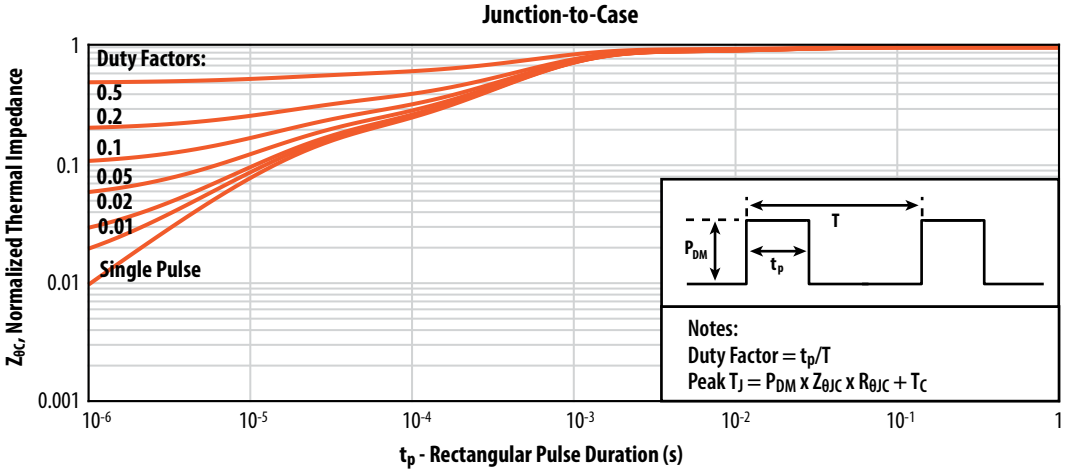


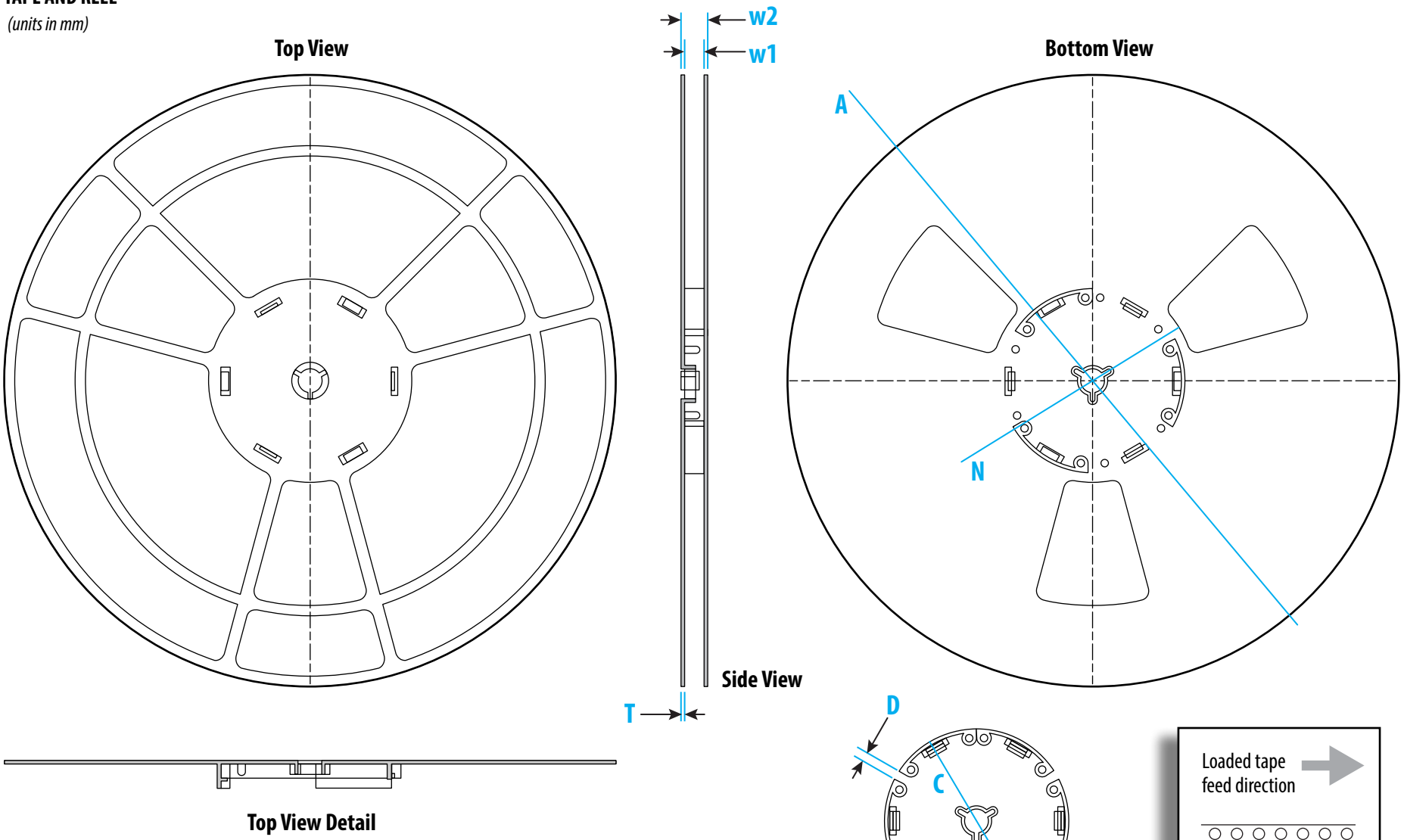
Figure 12: Transient Thermal Response Curves



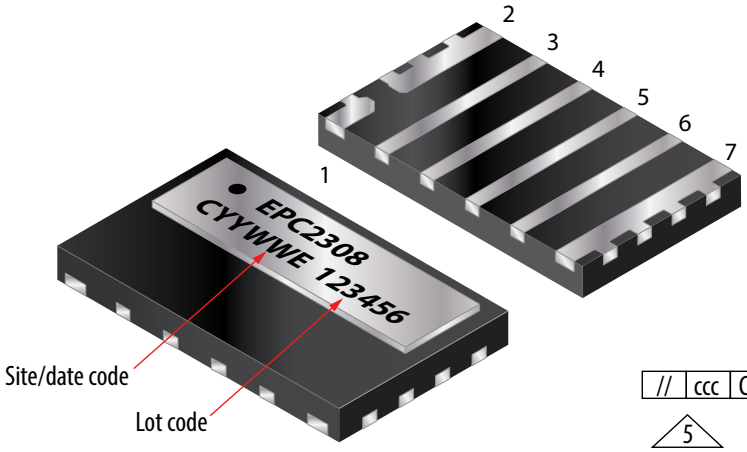


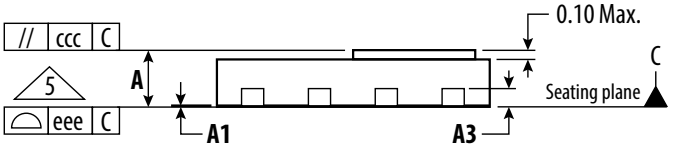
TAPE AND REEL

(units in mm)

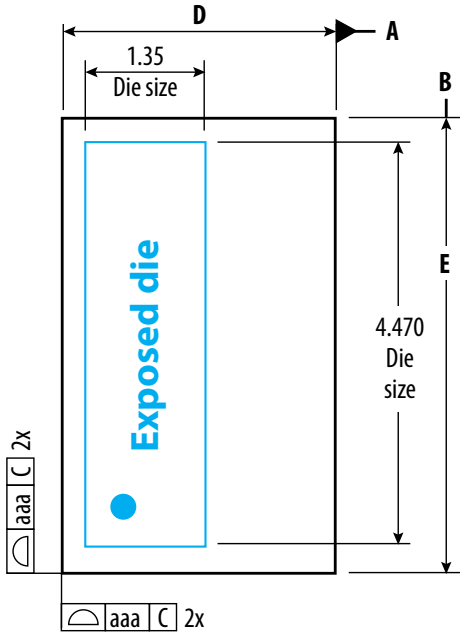


Type	A	N	C	D	w1	w2	T
8MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	8.4+1.5	14.4	2.1±0.5
12MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	12.4+1.5	18.4	2.1±0.5
16MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	16.4+1.5	22.4	2.1±0.5
24MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	24.4+1.5	30.4	2.1±0.5

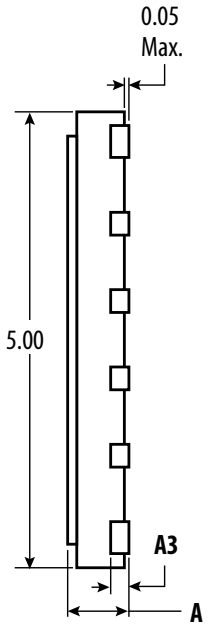




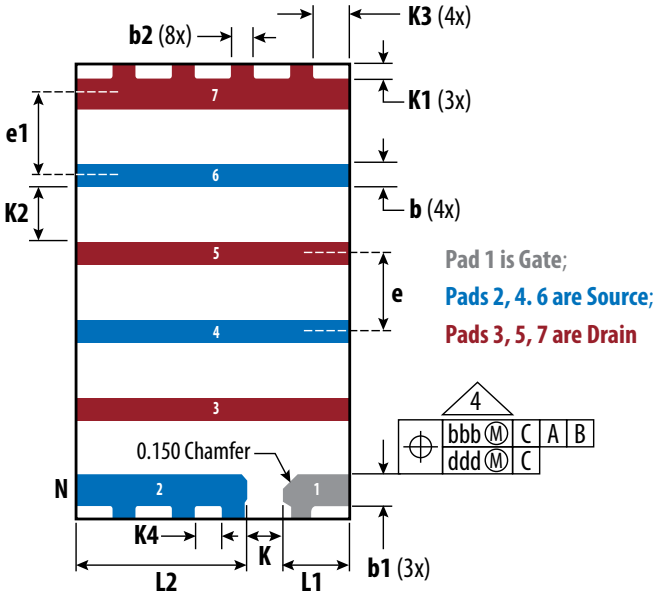
Side View 2



Top View



Side View 1



Bottom View

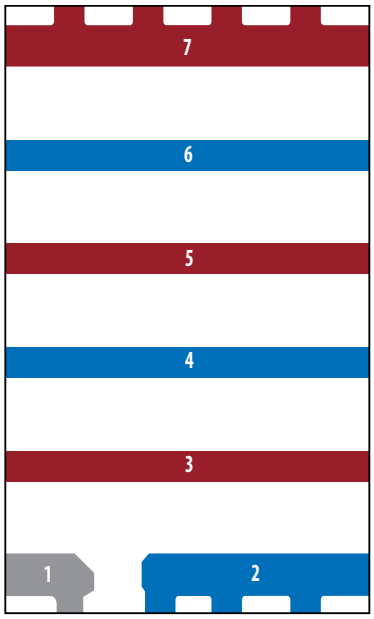
SYMBOL	Dimension (mm)			
	MIN	Nominal	MAX	Note
A	0.60	0.65	0.70	
A1	0.00	0.02	0.05	
A3		0.20 Ref		
b	0.20	0.25	0.30	4
b1	0.30	0.35	0.40	4
b2	0.20	0.25	0.30	4
D		3.00 BSC		
E		5.00 BSC		
e		0.85 BSC		
e1		0.90 BSC		
L1	0.625	0.725	0.825	
L2	1.775	1.875	1.975	

SYMBOL	Dimension (mm)			
	MIN	Nominal	MAX	Note
K	0.35	0.40	0.45	
K1	0.10	0.15	0.20	
K2	0.55	0.60	0.65	
K3	0.35	0.40	0.45	
K4	0.25	0.30	0.35	
aaa		0.05		
bbb		0.10		
ccc		0.10		
ddd		0.05		
eee		0.08		
N		15		3
NE		6		

Notes:

1. Dimensioning and tolerancing conform to ASME Y14.5-2009
2. All dimensions are in millimeters
3. **N** is the total number of terminals
4. Dimension **b** applies to the metallized terminal. If the terminal has a radius on the other end of it, dimension **b** should not be measured in that radius area.
5. Coplanarity applies to the terminals and all the other bottom surface metallization.

TRANSPARENT VIEW

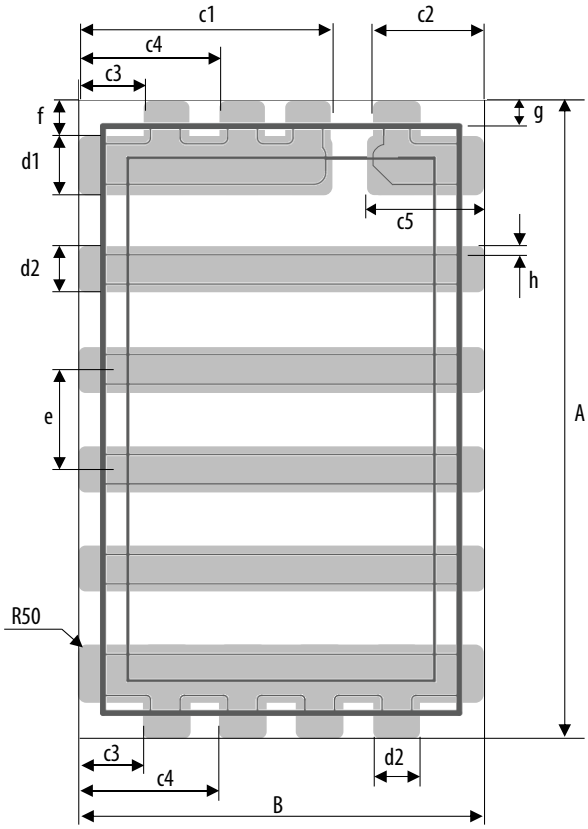


Transparent Top View

PIN	DESCRIPTION
1	Gate
2	Source
3	Drain
4	Source
5	Drain
6	Source
7	Drain

RECOMMENDED LAND PATTERN

(units in mm)



Land pattern is solder mask defined.

It is recommended to have on-Cu trace PCB vias.

DIM	Nominal
A	5.4
B	3.4
c1	2.1
c2	0.90
c3	0.55
c4	1.20
c5	0.975
d1	0.45
d2	0.35
e	0.85
f	0.30
g	0.2
h	0.05

Additional resources available:

- Assembly resources – https://epc-co.com/epc/Portals/0/epc/documents/product-training/Appnote_GaNassembly.pdf
- Library of Altium footprints for production FETs and ICs – https://epc-co.com/epc/documents/altium-files/EPC%20Altium%20Library.zip (for preliminary device Altium footprints, contact EPC)

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